

Temperature Measurement: Comparing Four Sensor Types

Thermocouple, RTD, thermistor, and infrared thermometer performance, calibration, and transient response

Author	Luis Ramirez	Performed	Spring 2026
Course	ME 343-004 Mechanical Lab I, NJIT	Instructor	Prof. B. S. Mani · TA: Saba Mirahsani
Team	Team 4: Luis Ramirez, Samuel Cohen, Edwin Orteiz, Abhin Ghandi, Ramid Brown	Equipment	Type K thermocouples, platinum RTD, thermistor, infrared thermometer, thermocouple module, multimeter, Vernier LabQuest Mini, hot plate

PROJECT SUMMARY

This experiment evaluated four temperature sensors — a Type K thermocouple, a platinum RTD, a thermistor, and an infrared thermometer — using ice, room-temperature, boiling, and mixed water baths as reference conditions. The thermocouple output was linear, with a measured sensitivity of 40.4 $\mu\text{V}/^\circ\text{C}$ that closely matches the standard Type K value of about 41 $\mu\text{V}/^\circ\text{C}$. The thermistor showed the expected steep, nonlinear drop in resistance with temperature, and fitting the data gives a material constant β of roughly 3,800 K. Transient tests showed that an open (exposed-junction) thermocouple responds almost instantly, while an enclosed probe takes several tenths of a second, highlighting how sensor construction governs response time.

1. Introduction

Accurate temperature measurement directly affects system performance, safety, and efficiency in many engineering applications. Temperature sensors operate on different physical principles, which gives each a different accuracy, sensitivity, and response speed. This experiment compared four sensor types under steady and changing conditions to understand those trade-offs and when each is the right choice.

2. Theory

Table 1. Sensor principles

Sensor	Operating principle	Governing relationship
Thermocouple	Two dissimilar metals produce a voltage proportional to the temperature difference between the measuring and reference junctions (Seebeck effect)	$E \approx S (T - T_{ref})$
RTD	Resistance of a pure metal (platinum) rises almost linearly with temperature	$R(T) = R_0(1 + AT + BT^2) \approx R_0[1 + \alpha(T - T_0)]$
Thermistor	Resistance of a semiconductor drops steeply and nonlinearly with temperature	$R(T) = R_0 \exp[\beta(1/T - 1/T_0)]$
Infrared	Measures thermal radiation emitted by a surface, with no contact	Depends on surface emissivity

For transient response, a sensor behaves as a lumped first-order system: $(T - T_\infty)/(T_0 - T_\infty) = e^{-t/\tau}$, where the time constant $\tau = mc/(hA)$ depends on the sensor’s mass, specific heat, surface area, and the heat transfer

coefficient. After one time constant, the sensor has covered 63.2% of a temperature step.

3. Procedure

The experiment had three parts. **Steady-state calibration:** each sensor was placed in ice water, room-temperature water, boiling water, and two mixed baths, with the thermocouple and signal conditioning module used as the reference. Thermocouple voltage was read directly on a multimeter, RTD and thermistor resistance were recorded, and the infrared thermometer was aimed at each sample. **Self-heating:** a Wheatstone bridge was used to study how excitation voltage heats the thermistor and shifts its reading. **Transient response:** a Vernier thermocouple connected to a LabQuest Mini was moved quickly between hot and cold baths while temperature was logged, for both open and enclosed probe types.



Figure 1. Water baths on the hot plate



Figure 2. Infrared thermometer measurement



Figure 3. Thermocouple voltage on the multimeter

4. Results

Table 2. Thermocouple results

Bath	Thermocouple reading (°C)	TC output (mV)	Vernier probe (°C)
Cold	8.3	-0.786	5
Room temperature	24.4	0.065	22
Hot + room temp.	47.4	0.830	45
Hot + cold	51.1	0.961	50
Hot (boiling)	100.0	3.000	102

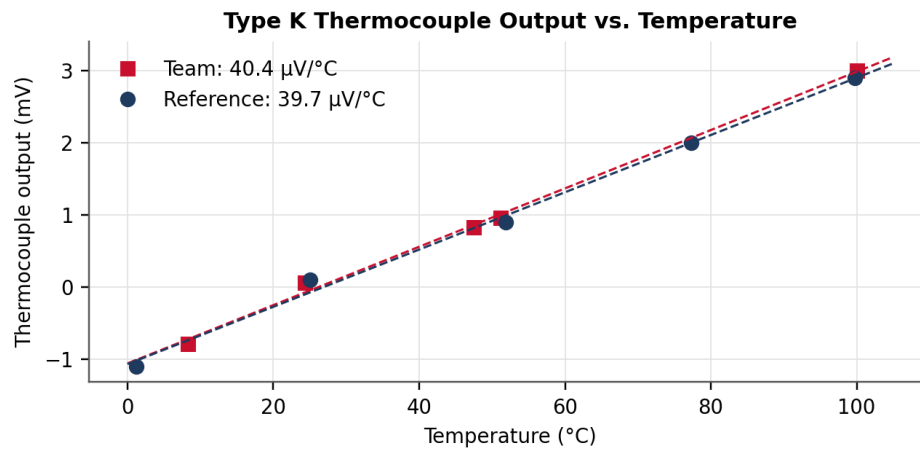


Figure 4. Thermocouple output vs. temperature, team data and lab reference data.

The output is linear ($R^2 = 0.997$) with a slope of $40.4 \mu\text{V}/^\circ\text{C}$, in line with the reference data ($39.7 \mu\text{V}/^\circ\text{C}$) and the standard Type K sensitivity. The fitted line crosses zero at about 26°C , which shows the reference junction was sitting near room temperature rather than at 0°C — the reason a thermocouple module applies cold-junction compensation.

Table 3. Thermistor results

Bath	Temperature (K)	Thermistor resistance (Ω)
Cold	278	5,445
Room temperature	295	2,385
Hot + cold	318	300
Hot + room temp.	323	342
Hot (boiling)	375	155

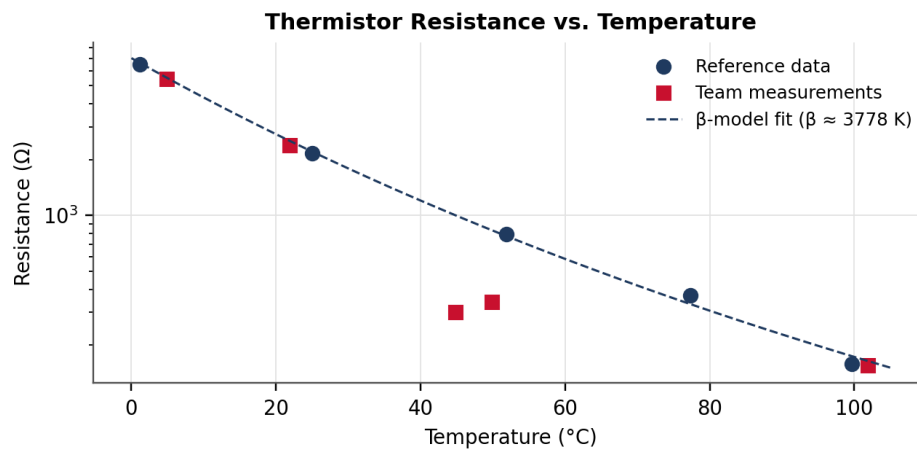


Figure 5. Thermistor resistance vs. temperature (log scale) with a β -model fit to the reference data.

Resistance fell by a factor of about 35 between 5°C and 102°C . Fitting the β model gives $\beta \approx 3,780 \text{ K}$ from the reference data and about $3,820 \text{ K}$ from the team's cold and boiling points, which is a typical value for an NTC thermistor. The two mixed-bath readings (300 and 342Ω) sit below the curve and are out of order, suggesting the thermistor was read before those baths had settled.

Transient response

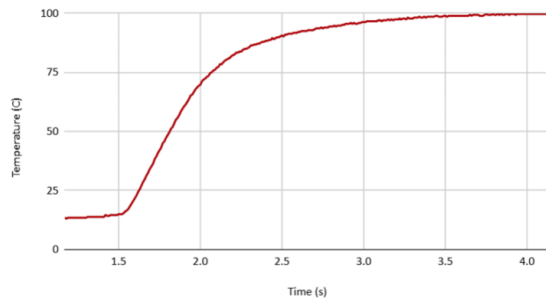


Figure 6. Cold to hot, enclosed probe

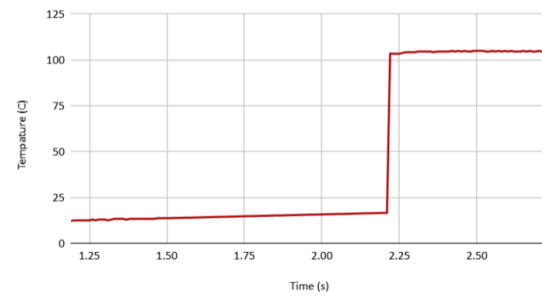


Figure 7. Cold to hot, open probe

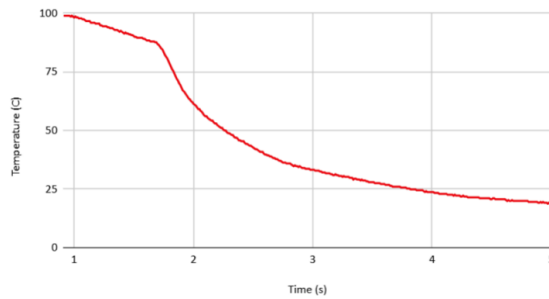


Figure 8. Hot to cold, enclosed probe

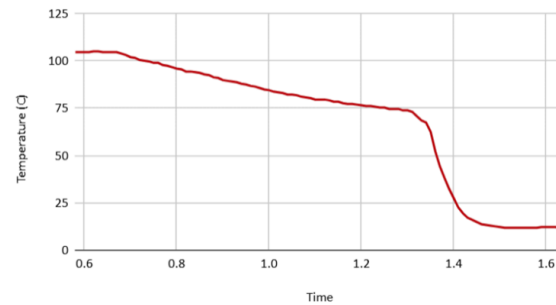


Figure 9. Hot to cold, open probe

Table 4. Time constants estimated from the graphs (time to cover 63.2% of the step)

Test	Probe	Approx. time constant τ
Cold $\rightarrow$ hot	Enclosed	≈ 0.4 s
Cold $\rightarrow$ hot	Open	< 0.05 s (near-instant step)
Hot $\rightarrow$ cold	Enclosed	≈ 0.8 s
Hot $\rightarrow$ cold	Open	≈ 0.07 s

5. Discussion

Steady-state behavior. The thermocouple was linear across the full range and agreed closely with the Vernier probe. The RTD's resistance increased with temperature along a slightly curved, quadratic trend, while the thermistor's resistance dropped sharply and nonlinearly. That steep curve gives the thermistor much higher sensitivity than the RTD, but only over a narrower range and at the cost of needing a nonlinear calibration. The infrared thermometer offered fast, non-contact readings but depends on the surface's emissivity.

Bath stability. The room-temperature bath was the most stable reference. The ice bath drifted as ice melted, and the boiling bath fluctuated with continuous heating and bubbling, which contributed small differences between sensors at the temperature extremes.

Transient response. Probe construction had a dramatic effect. The exposed-junction (open) thermocouple has very little mass, so it jumped across the step in a single sample, while the enclosed probe's sheath added thermal mass and slowed the response to a time constant of roughly half a second. In the hot-to-cold tests, the gradual decline before the sharp drop is the probe cooling in air while it was moved between baths. Cooling in the enclosed case was also slower than heating, consistent with a lower heat transfer coefficient as the water around the probe changed.

6. Conclusion

Each sensor responded according to its operating principle. The thermocouple was linear with a sensitivity matching Type K specifications, the RTD rose steadily along a slightly curved trend, and the thermistor fell steeply and nonlinearly with a β of about 3,800 K. Transient tests showed that sensor construction controls response time, with open probes responding roughly ten times faster than enclosed ones. The best sensor therefore depends on the application: thermocouples for wide range and speed, RTDs for accuracy and stability, thermistors for high sensitivity over a narrow range, and infrared for non-contact measurement.

References

- Holman, J. P. (2012). *Experimental methods for engineers* (8th ed.). McGraw-Hill.
- Zhu, C. (2009). *ME 343 laboratory introduction*. New Jersey Institute of Technology.